

γ -Glutamyl Transferase (GGT)

Test Details

METHODOLOGY

Kinetic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

A biliary enzyme that is especially useful in the diagnosis of obstructive jaundice, intrahepatic cholestasis, and pancreatitis.¹ GT is more responsive to biliary obstruction than are aspartate aminotransferase (AST) (SGOT) and alanine aminotransferase (ALT) (SGPT).

Increased in hepatoma and carcinoma of pancreas. Useful in diagnosis of metastatic carcinoma in the liver. Increasing levels in carcinoma patients relate to tumor progression, and diminishing levels to response to treatment.² CEA, alkaline phosphatase, and GT used together are useful markers for hepatic metastasis from breast and colon primaries. GT is elevated in some instances of seminoma.

Useful in diagnosis of chronic alcoholic liver disease, but some heavy drinkers do not have GT increases. Serial determinations of serum GT, AST, and ALT levels can distinguish recovering alcoholics who resume drinking from those who remain abstinent.^{3,4} Increase in body mass is positively correlated with increased GT levels.⁵ With MCV of red cells, GT is useful as a test for alcoholism.

GT is the test for cholestasis during or immediately following pregnancy. Commonly elevated in cirrhosis and hepatitis. The transaminases, AST and ALT rise higher in acute viral hepatitis; these tests with GT and other parameters are best used together in work-up of liver disease.

Increased in systemic lupus erythematosus.² Very high levels are common in primary biliary cirrhosis. High GT is found in infants with biliary atresia. It is increased with hyperthyroidism and decreased in those with hypothyroidism.⁶ GT is comparable in many ways to two other biliary tests, LAP and 5' nucleotidase. In some cases, five tests (including alkaline phosphatase and bilirubin) are necessary to evaluate the biliary tract. GT usually is the most sensitive.

In **ascitic fluid**, very high GT is said to suggest hepatoma as opposed to cirrhosis or liver metastases.

Limitations

Acetaminophen toxicity has been reported to cause an in vivo increase. The combination of high alkaline phosphatase and normal GT does not rule out liver disease completely. Activity is not significantly increased in sera of patients with lymphoma (unless there is hepatic involvement by the lymphoma). Baden et al concluded that as a preoperative test for metastasis with colorectal carcinoma, GT is unsatisfactory.⁷ As part of a evaluation for carcinoma patients, 19% of GT results from patients with progressive disease were not abnormal, and 4% of values from patients without evidence of tumor were high.²

Custom Additional Information

GT is helpful to work up elevated alkaline phosphatase values. GT is a biliary excretory enzyme which is more specific for hepatic disease than is alkaline phosphatase. It is normal in most instances of renal failure.⁸ GT has no origin in bone or placenta, unlike alkaline phosphatase, and age beyond infancy does not influence GT levels. Activity of GT is highest in obstructive liver disease. It is commonly elevated in patients with infectious mononucleosis. When GT and alkaline phosphatase are both high, but one is disproportionately elevated, suspect the possibility of drug-induced cholestasis (including alcoholism if it is GT which is much higher). GT, postprandial glucose, and triglyceride bear some correlation in certain groups of patients, including alcoholism and diabetes mellitus. Treatment of hypertriglyceridemia may also lead to decreased GT. **GT is normal** in normal children, adolescents, and in pregnant women. Unlike AST, it is not elevated in skeletal muscle disease.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

- Male: 0–65 IU/L
- Female: 0–60 IU/L

Storage Instructions

Maintain specimen at room temperature.

Patient Preparation

The patient should fast for eight hours prior to collection of the specimen. Since there are false elevations in patients on phenytoin and phenobarbital, such patients would be better served with orders for one of the alternate tests – leucine aminopeptidase (LAP) or 5' nucleotidase.

Causes for Rejection

Gross hemolysis; improper labeling; gross lipemia

Footnotes

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2. Sahm DF, Murray JL, Munson PL, Nordquist RE, Lerner MP. Gamma-glutamyltranspeptidase levels as an aid in the management of human cancer. *Cancer.* 1983 Nov 1; 52(9):1673-1678. PubMed 6137275 (<http://www.ncbi.nlm.nih.gov/pubmed/6137275>)
3. Irwin M, Baird S, Smith TL, Schuckit M. Use of laboratory tests to monitor heavy drinking by alcoholic men discharged from a treatment program. *Am J Psychiatry.* 1988 May; 145(5):595-599. PubMed 2895984 (<http://www.ncbi.nlm.nih.gov/pubmed/2895984>)
4. Frimpong NA, Lapp JA. Effects of moderate alcohol intake in fixed or variable amounts on concentration of serum lipids and liver enzymes in healthy young men. *Am J Clin Nutr.* 1989 Nov; 50(5):987-991. PubMed 2573268 (<http://www.ncbi.nlm.nih.gov/pubmed/2573268>)
5. Robinson D, Whitehead TP. Effect of body mass and other factors on serum liver enzyme levels in men attending for well population screening. *Ann Clin Biochem.* 1989 Sep; 26(Pt 5):393-400. PubMed 2573311 (<http://www.ncbi.nlm.nih.gov/pubmed/2573311>)
6. Schaffner F. Tests related to the liver. *Gastroenterology.* 1985; 410-426.
7. Baden H, Andersen B, Augustenborg G, Hanel HK. Diagnostic value of gamma-glutamyl transpeptidase and alkaline phosphatase in liver metastases. *Surg Gynecol Obstet.* 1971 Nov; 133(5):769-773. PubMed 5111112 (<http://www.ncbi.nlm.nih.gov/pubmed/5111112>)
8. Lum G, Gambino SR. Serum gamma-glutamyl transpeptidase activity as an indicator of disease of liver, pancreas, or bone. *Clin Chem.* 1972; 18(4):358-362. PubMed 5012259 (<http://www.ncbi.nlm.nih.gov/pubmed/5012259>)